

**Option #2: NLP Chatbot Project Draft**

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## **Option #2: NLP Chatbot Project Draft**

For the final portfolio project, the selected option is Option #2, the NLP Chatbot Project. The proposed system is a closed-domain enterprise knowledge-support chatbot designed to help employees retrieve accurate answers from a curated company knowledge base. The initial use case will focus on common operational questions such as policy lookup, process guidance, onboarding support, and frequently requested internal procedures. This direction is intentionally practical: many organizations have information stored across manuals, standard operating procedures, training documents, and frequently asked question lists, but employees often struggle to locate the right answer quickly. The chatbot's goal will be to interpret a user's natural-language question, identify the most relevant knowledge-base entry, and return a clear response with enough context for the employee to act.

The chatbot will demonstrate adaptive control in an uncertain environment by treating each user question as an uncertain input condition. The system will not assume that users know the exact policy name, document title, or vocabulary used in the knowledge base. Instead, it will adapt its response strategy based on the similarity between the user's question and the available training entries. When confidence is high, the bot will return the best-matching answer. When confidence is moderate, it will return a cautious response and may suggest the closest matching topic. When confidence is low, it will ask the user to rephrase or narrow the question rather than inventing an unsupported answer. This confidence-based response control is important because a business chatbot should avoid nonsense responses and should make uncertainty visible to the user.

### **Technical Plan**

The first version will be a retrieval-based chatbot rather than an open-domain generative chatbot. The planned implementation will use Python, pandas for loading the knowledge-base dataset, and scikit-learn for text vectorization and similarity scoring. The core model will use term frequency-inverse document frequency (TF-IDF) features and cosine

similarity to compare the user's input with stored questions and answers. TF-IDF is a reasonable first model because it is explainable, fast, and well suited for matching user queries against a controlled document collection (Manning et al., 2008). Scikit-learn is also appropriate for this coursework prototype because it provides a stable Python machine learning interface and includes text feature extraction tools such as `TfidfVectorizer` (Pedregosa et al., 2011; scikit-learn developers, 2025).

The program structure will include a data folder with a CSV knowledge base, a preprocessing module, a retrieval model module, a chatbot interface, and a small testing script. The dataset will contain fields such as topic, sample question, approved answer, source category, and optional keywords. The preprocessing step will normalize case, remove unnecessary punctuation, and prepare text for vectorization. The retrieval model will fit the TF-IDF vectorizer on the knowledge-base entries and rank candidate answers using cosine similarity. The chatbot interface will accept user input from the command line or a lightweight web interface, display the selected answer, and show a confidence score for testing. Initial hyperparameters will include unigram and bigram features, a maximum document-frequency threshold to reduce overly common terms, and a minimum similarity threshold for fallback behavior.

The final version may compare this baseline with a sentence-embedding approach or transformer-based model if time and dependency setup permit. Transformer models are important in modern NLP because attention-based architectures improved the ability of models to represent language context (Vaswani et al., 2017). However, for the final course deliverable, the safest design is a closed-domain retrieval system that can be tested with known expected answers. The project will be evaluated by running sample employee questions through the chatbot, recording whether the retrieved answers are relevant, and identifying where additional training examples or better threshold tuning are needed. This approach creates a defensible prototype for later enterprise use.

## References

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